VWO Login Dashboard Test Plan

Project: VWO (Visual Website Optimizer) Login Dashboard (app.vwo.com) **Document Type:** Strategic Test Plan **Version:** 1.0

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1. Introduction

Overview:
This Test Plan outlines the strategic approach for validating the VWO login dashboard (app.vwo.com). VWO is a leading digital experience optimization platform utilized by over 4,000 brands across 90 countries. The login dashboard acts as the critical, secure entry point to VWO's comprehensive suite of A/B testing, conversion rate optimization (CRO), and analytics tools.

Goals & Context:
The primary testing goal is to ensure the delivered login experience is highly secure, intuitive, and efficient. It must flawlessly connect users—ranging from growing businesses to large enterprises—to the optimization platform. Key objectives include validating enterprise-grade security standards, verifying the elimination of login friction to boost adoption, ensuring strict compliance adherence, and confirming a seamless onboarding flow for new users.

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2. Scope (Tested & Not Tested)

In-Scope (Tested): The following core features and technical implementations are explicitly within the testing scope, categorized by development phases outlined in the PRD:

- **Authentication Flow:** Primary email/password authentication, secure token generation, session management, multi-factor authentication (2FA), and enterprise Single Sign-On (SSO - SAML, OAuth, Social Logins).
- **Input & Feedback Output:** Real-time form validation (on blur), programmatic email format verification, password strength indicators, auto-focus behavior, and clear error messaging for authentication failures.
- **Account Recovery:** Password reset and recovery flows.
- **Accessibility & UX:** Responsive (mobile-optimized) layouts, theme support (Light and Dark Mode), screen reader (ARIA) compatibility, High Contrast Mode, global keyboard navigation, and WCAG 2.1 AA compliance mechanisms.
- **Performance:** Validating sub-2-second page load times, image compression, minified assets, CDN integration, and backend scalability for thousands of simultaneous requests.
- **Security & Compliance:** End-to-end data encryption (HTTPS/TLS), brute-force protection (rate limiting), secure password hashing verification, and session timeout/hijacking prevention. Validation of GDPR/CCPA alignment.
- **Platform Integration:** Transition mechanisms to the primary VWO dashboard, analytics tracking integration for login telemetry, and customer support integration links.

Out-of-Scope (Not Tested):

- **Future Enhancements:** Biometric authentication (fingerprint/facial recognition), Adaptive/Risk-based authentication, and Progressive Web App (PWA) functionality are not tested, as they are slated for future releases.
- **Post-Login Platform Tools:** Validating the A/B testing engine, complex analytics models, or internal platform rules beyond the initial dashboard transition is out of scope.

- ***Reasoning:*** Explicitly omitted to maintain strict focus on the authentication boundary and adhere to the specified PRD Phase 1-3 roadmap.

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3. Test Strategy/Approach

Methodology: The testing methodology will follow a risk-based, pragmatic strategy aligned with the three delineated development phases (Phase 1: Core Authentication; Phase 2: Enhanced UX; Phase 3: Enterprise Features).

Types of Testing:

- **Functional Testing:** Validation of state transitions, form verifications, authentication gateways, SSO assertions, and exception recovery via automated and exploratory paths.
- **Performance & Load Testing:** Comprehensive simulations to assess concurrent load stability (thousands of simultaneous attempts), auto-scaling efficacy, and strict adherence to the sub-2-second load SLA.
- **Security & Penetration Testing:** Rigorous evaluation against OWASP authentication guidelines, active penetration tests, session hijacking simulations, and validation of payload encryptions.
- **Accessibility Testing:** Verification of inclusive design principles enforcing WCAG 2.1 AA standards, utilizing assistive technology emulators, screen readers, and programmatic accessibility audits.
- **A/B & UX Optimization Testing:** A continuous optimization loop leveraging continuous testing of the login experience.

Tools to be Used:

- **Test Automation:** Playwright (Mandatory for End-to-End and functional automation workflows).
- **Analytics & A/B:** VWO's Internal Platform (For continuous A/B testing and user behavior analytics).
- *Other Tools (e.g., Performance, Security):* **[NOT SPECIFIED]**

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4. Test Environment

Hardware/Software Components:

- Compatible Desktop & Mobile Devices parameters: **[NOT SPECIFIED]** (General requirement is responsive/mobile-optimized).
- OS/Browser Matrices: **[NOT SPECIFIED]**

Networks:

- Standard consumer network configurations for <2s load validations.
- Multi-region deployment connections for optimal global performance validation.
- CDN environments.

Data & Configuration Environments:

- Other environment parameters: **[NOT SPECIFIED]**

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5. Test Criteria

Entry Criteria:

- **[NOT SPECIFIED]**

Exit / Pass-Fail Criteria: Testing must prove the system meets the definitively established Key Performance Indicators (KPIs):

- **Functional / Reliability:** Attain a 95%+ successful login authentication rate.
- **Performance & Availability:** Maintain sub-2-second login page loading. Validate infrastructure achieves 99.9% holistic uptime under load.
- **Security:** Achieve zero (0) successful brute force attacks or unauthorized access events. Record zero (0) unauthorized session hijackings. Exhibit 100% adherence to security audit and compliance policies.
- **Business / User Satisfaction:** Achieve a 90%+ user satisfaction score specifically regarding the login experience. Demonstrate a 20% reduction in login-related direct support tickets.

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6. Resources & Roles

Team Member Roles & Responsibilities:

- **[NOT SPECIFIED]**

Training Needs:

- **[NOT SPECIFIED]**

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7. Schedule & Milestones

Key Dates & Deadlines:

- **[NOT SPECIFIED]**

Project Milestones Sequence: While exact deployment dates are pending, milestone sequencing explicitly correlates with the product deployment phases:

- *Milestone 1 (Q1):* Phase 1 rollout (Core Authentication features).
- *Milestone 2 (Q2):* Phase 2 rollout (Enhanced UI/UX, responsive layouts, and standard accessibility).
- *Milestone 3 (Q3):* Phase 3 rollout (Enterprise SSO, strict security additions, and complex telemetry).

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8. Risk & Contingency

Security Risks:

- *Risk:* Unauthorized access, active vulnerability exploits, or brute force intrusions.
- *Contingency / Mitigation Plan:* Execute regular security audits and rigorous active penetration testing. Deploy real-time security monitoring layers and alert mechanisms. Implement scheduled, recurring security patching and vulnerability assessments.

Performance Risks:

- *Risk:* System degradation during traffic spikes impacting the login pathway SLA.
- *Contingency / Mitigation Plan:* Conduct comprehensive load testing covering diverse load distribution conditions. Enforce real-time performance application monitoring schemas. Rely on responsive auto-scaling infrastructure topologies.

Compliance Risks:

- *Risk:* Failure to adhere strictly to regulatory data privacy and inclusive design mandates.

- *Contingency / Mitigation Plan:* Map architectures explicitly to GDPR, CCPA, and enterprise security governance structures. Standardize workflows relying on established OWASP parameters. Implement continuous accessibility analysis to protect WCAG 2.1 AA mandates.

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9. Deliverables

- **[NOT SPECIFIED]** (Awaiting formal reporting requirement protocols, beyond standard security audit compliance data and PRD artifacts).